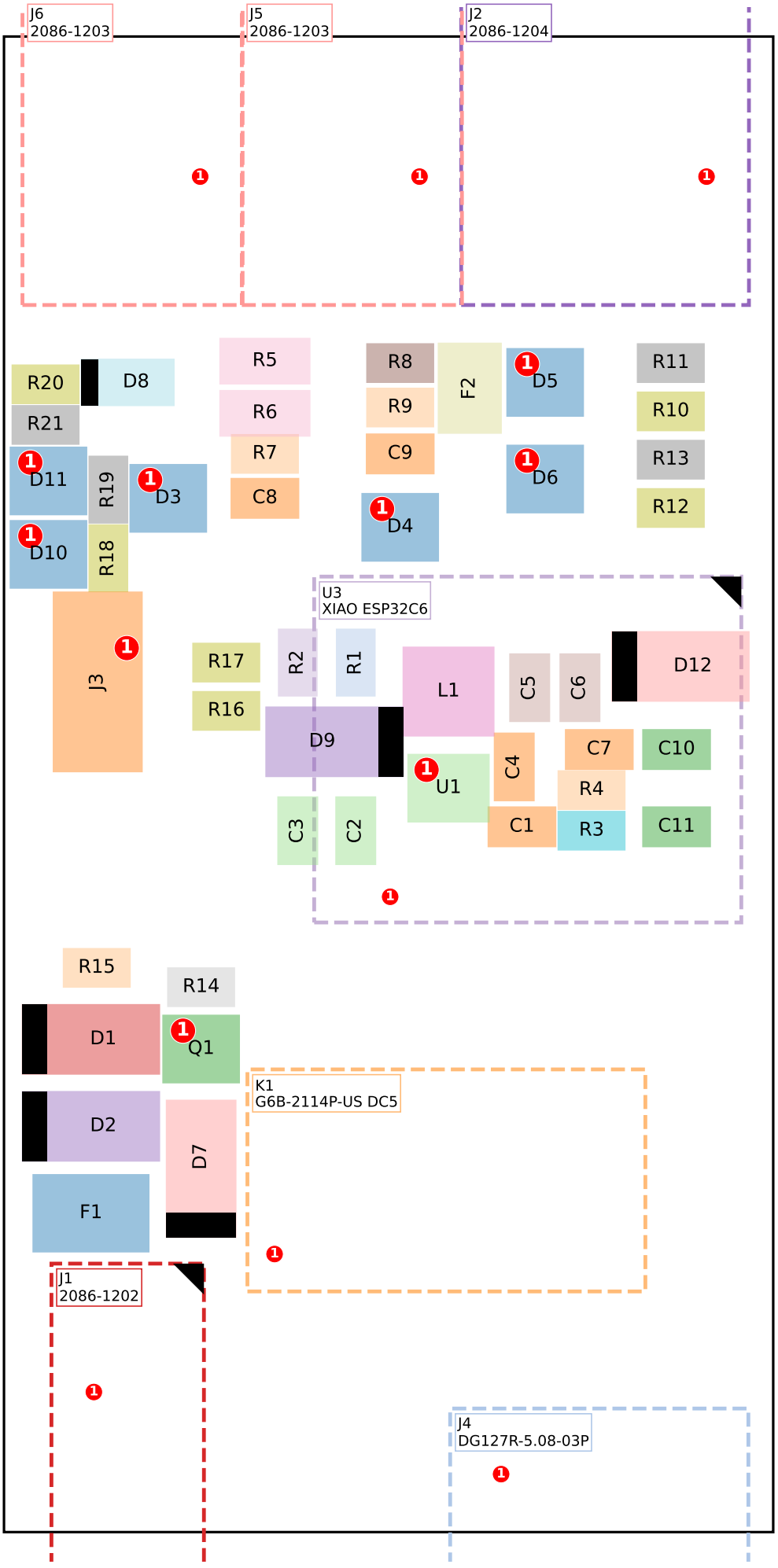
# main board: TOP side

black bar = diode cathode;  red + = capacitor +;  red 1 = pin 1;  black corner = hand-fitted;  same colour = same value (values page)



# main board: parts by value

| | | | | |
|---|---|---|---|---|
| **1.1A (1812)** | x1 | top | - | F1<br>1812L110/33MR C142747 |
| **100k (0805)** | x1 | top | - | R1<br>RC0805FR-07100KL C96346 |
| **100n 50V (0805)** | x5 | top | - | C1, C4, C7, C8, C9<br>CC0805KRX7R9BB104 C49678 |
| **10k (0805)** | x4 | top | - | R4, R7, R9, R15<br>0805W8F1002T5E C17414 |
| **10u 25V (0805)** | x2 | top | - | C10, C11<br>CL21A106KAYNNNE C15850 |
| **10u 50V (0805)** | x2 | top | - | C2, C3<br>GRM21BR61H106KE43L C440198 |
| **2086-1202 (TerminalBlock)** | x1 | top | pin 1 | J1<br>2086-1202/700-000/997-604 |
| **2086-1203 (TerminalBlock)** | x2 | top | pin 1 | J5, J6<br>2086-1203/700-000/997-605 C3309626 |
| **2086-1204 (TerminalBlock)** | x1 | top | pin 1 | J2<br>2086-1204/700-000/997-605 C3309718 |
| **20k (0805)** | x1 | top | - | R2<br>0805W8F2002T5E C4328 |
| **22k (0805)** | x1 | top | - | R8<br>0805W8F2202T5E C17560 |
| **22u 25V (0805)** | x2 | top | - | C5, C6<br>CL21A226MAQNNNE C45783 |
| **22uH (L)** | x1 | top | - | L1<br>FNR4020S220MT C167838 |
| **240R (1206)** | x2 | top | - | R5, R6<br>1206W4F2400T5E C17961 |
| **330R (0805)** | x4 | top | - | R11, R13, R19, R21<br>0805W8F3300T5E C17630 |
| **470R (0805)** | x1 | top | - | R14<br>0805W8F4700T5E C17710 |
| **4k7 (0805)** | x6 | top | - | R10, R12, R16, R17, R18, R20<br>0805W8F4701T5E C17673 |
| **50mA (1210)** | x1 | top | - | F2<br>MF-USMF005-2 C118758 |
| **56k (0805)** | x1 | top | - | R3<br>0805W8F5602T5E C17756 |
| **5V6 (SOD-123)** | x1 | top | cathode bar | D8<br>MMSZ5V6T1G C236111 |
| **BAT54S (SOT-23)** | x6 | top | pin 1 | D3, D4, D5, D6, D10, D11<br>BAT54S,215 C47546 |
| **DG127R-5.08-03P (TerminalBlock)** | x1 | top | pin 1 | J4<br>DG127R-5.08-03P-14-00A(H) C699499 |
| **FPC-05F-8PH20 (FPC)** | x1 | top | pin 1 | J3<br>FPC-05F-8PH20 C2856797 |
| **G6B-2114P-US DC5 (Relay)** | x1 | top | pin 1 | K1<br>G6B-2114P-US DC5 C397217 |
| **MMBT3904 (SOT-23)** | x1 | top | pin 1 | Q1<br>MMBT3904(RANGE:100-300) C20526 |
| **SCT2601TVBR (TSOT-23-6)** | x1 | top | pin 1 | U1<br>SCT2601TVBR C2931912 |
| **SMAJ30A (SMA)** | x1 | top | cathode bar | D1<br>SMAJ30A C113965 |
| **SS34 (SMA)** | x2 | top | cathode bar | D7, D12<br>SS34 C8678 |
| **SS36 (SMA)** | x2 | top | cathode bar | D2, D9<br>SS36 C16015 |
| **XIAO ESP32C6 (XIAO)** | x1 | top | pin 1 | U3<br>102010636 |

# main board: key

## Marks on the drawing

Black bar across a diode: the cathode end. Put the band printed on the diode on this side.

Red +: the + pad of an electrolytic or tantalum capacitor.

Red dot with 1: pin 1 of an IC, transistor, LED, connector or module.

No mark: resistors, ceramic capacitors, inductors, fuses, crystals, pin strips and push buttons can go either way round. Bidirectional TVS diodes (part numbers ending in CA) have no polarity either.

Black top-right corner: hand-fitted part (not placed by JLCPCB), e.g. a socketed module or a terminal JLC has no stock of.

The bottom page is mirrored: it shows the board as you see it after flipping it over.

## Finding pin 1 on the real part

SOT-23 (3 legs): the single leg on its own side is pin 3; pins 1 and 2 are on the opposite side.

SOT-23-5 / TSOT-23-6 / SC-70 / SO / QFN: a dot or a bevelled edge on the body marks pin 1.

Addressable RGB LEDs: one corner of the body or lens is marked (a cut corner or a dot). Which pin it marks differs between makers: check the part's datasheet before soldering, it is easy to misread.

0805 LEDs: pin 1 is the cathode, usually a green line or a T on the underside.

Diodes: the band is the cathode.

Electrolytic capacitors: the stripe on the can marks the - side; the + pad is the other one.

Modules and connectors: match the pin-1 dot to the module's pin-1 marking or to the drawing in its datasheet; for modules on sockets, check the outline (e.g. USB connector towards the board edge).

## Tools

0805 parts and SOT-23: a 3-5x magnifier lamp or a 10x loupe to read markings and check joints.

SC-70, QFN and fine-pitch connectors: a USB microscope helps to see bridges and pin-1 marks.

Flux, fine solder (0.5 mm) and a chisel or fine conical tip; tweezers for placing.